

3. power Amplifiers

Introduction

* The power amplifiers are the amplifiers which raise the power levels of the signal.

* As a device which converts dc power to ac power and whose action is controlled by the dc i/p signal.

* The power amplifier does not amplify the power, in fact the power amplifier takes power from dc power supply connected to the circuit and converts it into ac signal power.

* The power amplifier is an essential part of every electronic system for example, it is used as last stage in the broadcast transmitters and radio receivers, television etc.

* Are also known as "large signal amplifiers" because they use large part of load line for their operation.

the ac i/p signal applied $V_{in} > V_T$.

V_T = Threshold voltage

Difference b/w voltage amplifiers and power amplifiers :-

(2)

Voltage amplifier	Power amplifier
1) Produces increased voltage in a high resistance load 2) Transistor used has a large current gain around 100db. 3) Δ/p resistance low as compared to o/p resistance. 4) physical size of transistor is very small (low or medium power transistor) 5) Normally uses RC coupling. 6) It amplifies both AF and RF signals. 7) Input applied $V_i < V_T$ 8) Transistor is operated in active region. So, there is no distortion.	Produces 1) Increased power in a low resistance load. - Has small current gain b/w 20-50db. Δ/p resistance high compared o/p resistance. - physical size is large. This is required because considerable amount of power is dissipated in the transistor and hence large surface area is required for heat dissipation. - uses Transformer coupling. - It amplifies AF signals. $V_p > V_T$ - Transistor operated in cut off or beyond cut off so there is some distortion.

Performance Measures of Power Amplifiers :-

(i) Circuit efficiency (or) overall conversion efficiency :-

The ability of the power amplifier to convert dc power to ac power is a measure of its effectiveness.

$$\% \eta = \frac{\text{Max. ac o/p power}}{\text{dc i/p power applied}} \times 100$$

* η lies b/w 25 to 90%.

(ii) Distortion :- The change of o/p wave shape from the i/p wave shape of amplifier. It occurs because in large signal amplifier the transmission operated in nonlinear region (cut off or beyond cut off region).

(iii) power dissipation capability :-

The ability of the power transistor to dissipate

heat is known as "power dissipation capability or power rating".

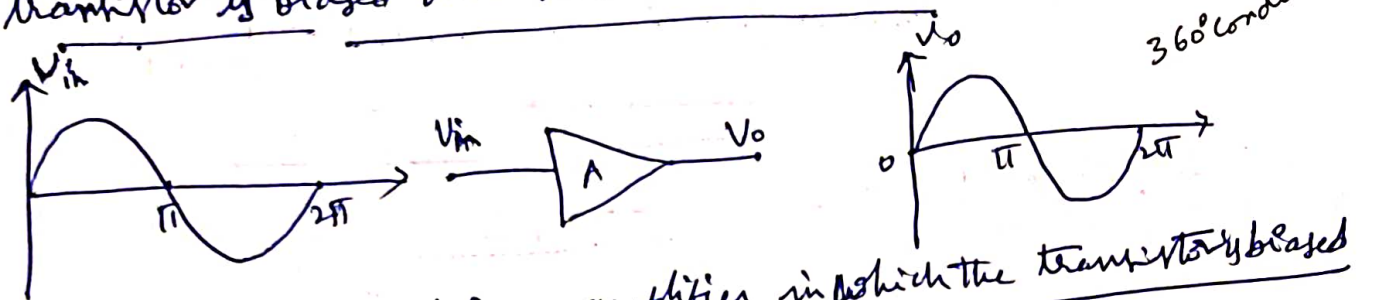
If heat generated is greater than power dissipation capability, then the transistor will burn, so a heat sink is connected to transistor.

Classification of Power Amplifiers :-

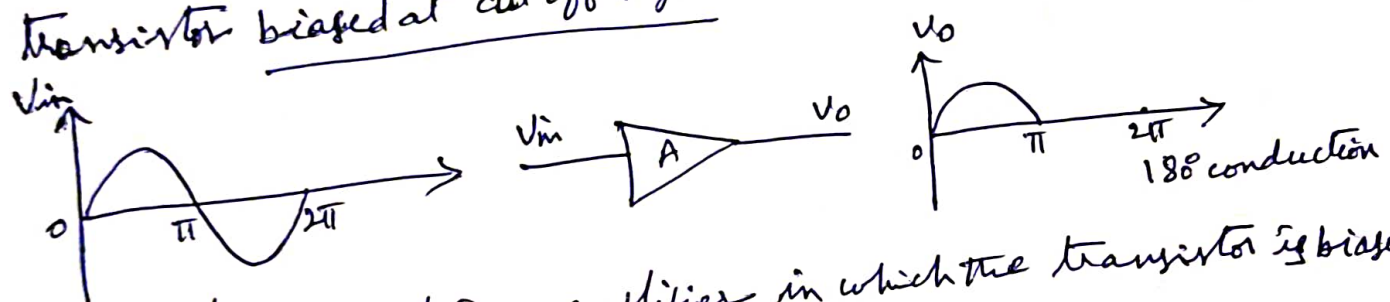
The transistor biasing condition, amplitude of i/p signal and the conduction period of transistor decides the classification of PA.

- i) class A
- ii) class B
- iii) class C
- iv) class D
- v) class AB
- vi) class S

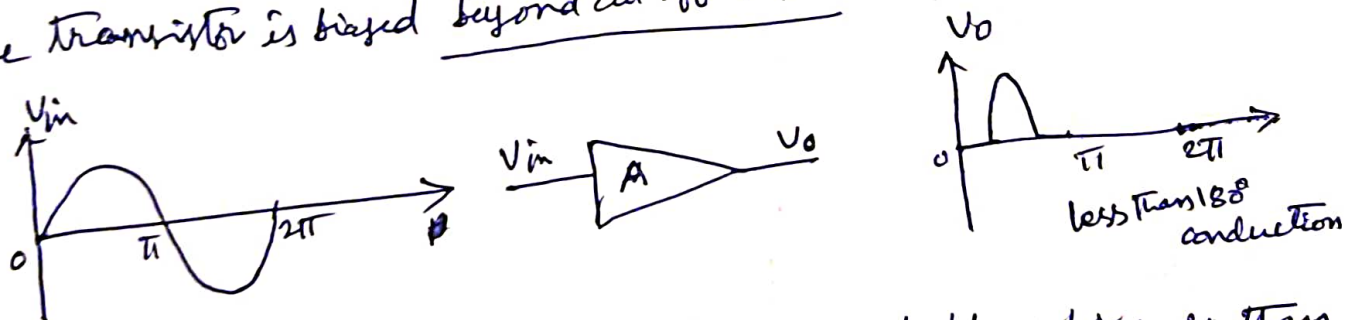
i) Class A Amplifier :- It is an amplifier in which the transistor is biased in such a way that the o/p current flows for the complete cycle of i/p and the transistor is biased well within active region.



ii) Class B power Amplifier :- It is an amplifier in which the transistor is biased such that the o/p current flows only for one half cycle of i/p signal. The transistor is biased at cut off region.



iii) Class C Amplifier :- It is an amplifier in which the transistor is biased such that o/p current flows for less than half cycle of i/p signal. The transistor is biased beyond cut off region.

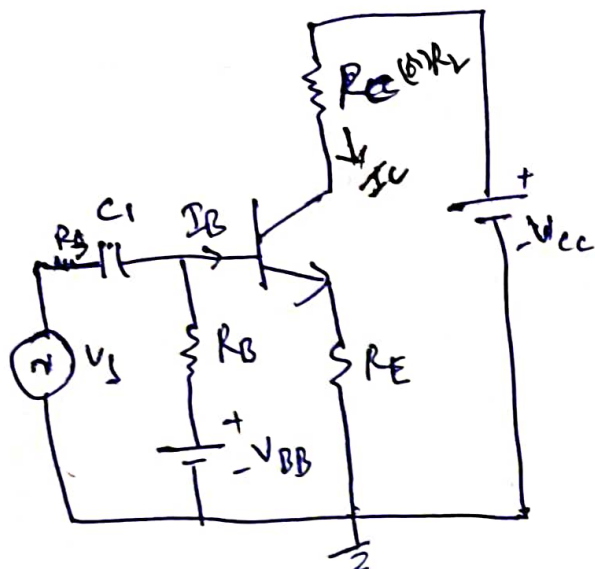


iv) In class AB amplifier the o/p current flows for more than half cycle but less than full cycle.

* class A and class B amplifiers are called Audio power amplifiers because they are capable of amplifying the signals in the range 20 Hz to 20 kHz.

Class-A power Amplifier

Construction :-



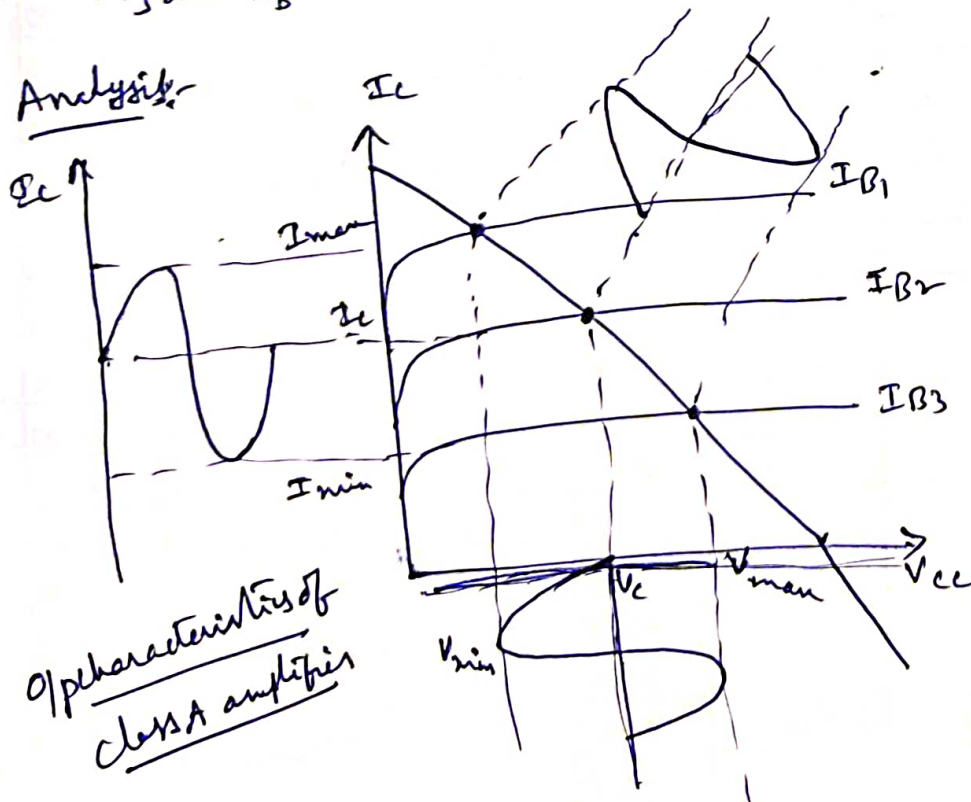
(5)
* It is an amplifier in which the o/p current flows for the complete ac cycle of the i/p signal. This is achieved by locating the Q point at the centre of load line or active Region.
class A amplifier is a CE amplifier and the o/p is directly coupled with load. Hence it is "direct coupled class A power amplifier".

* In order to operate the transistor exactly at midpoint of load line, two separate power supplies are used.

* The resistor R_B provides biasing voltage to base-emitter junction through V_{BB} and V_{CC} provides biasing for base-collector junction.

* The i/p signal is coupled to base via C_1 , it prevents loading effect between R_s and R_B . The resistor R_E acts as temperature stabilization resistor.

Analysis:-



o/p characteristics of class A amplifier

$$I_c = \frac{I_{max} - I_{min}}{2\sqrt{2}} = \frac{I_m}{\sqrt{2}} = \frac{I_{PP}}{2\sqrt{2}}$$

$$I_m = I_{PP}$$

* If i_b is sinusoidal then i_c is also sinusoidal.
* Under ideal conditions the nonlinear distortion is negligible and
o/p power $P = \frac{V_{CC} I_c}{2} = \frac{I_c^2 R_L}{2}$

where V_c and I_c are RMS values of o/p voltage and current respectively.

$$V_C = \frac{V_{PP}}{2\sqrt{2}} = \frac{V_m}{\sqrt{2}}$$

$$P_{oman} = V_C I_C = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} = \frac{I_m^2 R_L}{2} = \frac{V_m^2}{2R_L}$$

operation:- 1) when ac i/p signal is applied to the base, the transistor is biased in such a way that, the o/p current flows for the entire ac cycle.

2) In this case, the Q point is selected at the centre of load line so that it can operate in linear region and the o/p is 180° to i/p waveform.

3) when ac i/p signal is applied at the base of transistor, the o/p will vary from dc bias point. If i/p is made larger, the o/p will vary further i.e. varies from minimum to maximum.

4) If the large signals shift the Q point to cut off or saturation region and it provides amplitude distortion.

5) In this amplifier, even if i/p signal is absent, due to biasing voltage, the power is wasted in the form of heat across R_L and the power dissipation is more across R_L and efficiency is very low (25%).

6) To avoid this problem instead of using resistor as a load, transformer may be used which improves efficiency from 25% to 50%.

Power relation

$$V_{CC(p-p)} = V_{CC}, \quad I_{C(p-p)} = \frac{V_{CC}}{R_L}$$

$$P_{oman} = \frac{V_{CC}^2}{8R_L} = \frac{V_{CC}^2}{8R_L}$$

$$P_{ide} = V_{CC} I_{CQ} = V_{CC} \cdot \frac{V_{CC}}{2R_L} = \frac{V_{CC}^2}{2R_L}$$

$$\therefore \% \eta = \frac{P_{ac max} \times 100}{P_{ide}} = \frac{\frac{V_{CC}^2}{8R_L}}{\frac{V_{CC}^2}{2R_L}} \times 100 = 25\%$$

Advantages:- ① The circuit is simple and easy to design

⑦

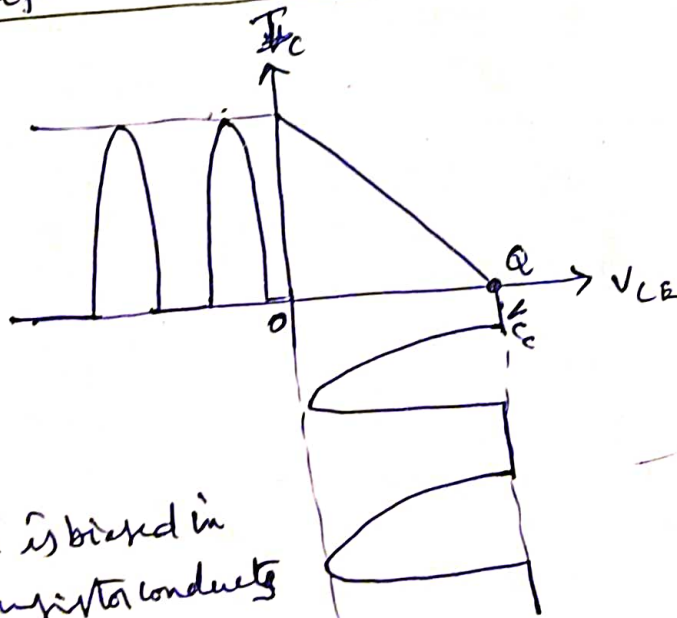
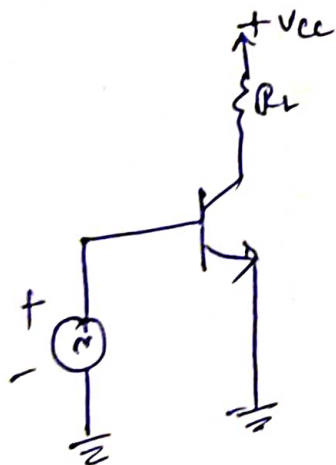
② The load is directly ^{connected} to collector circuit, thus the

O/p coupling element is not required (circuit is cheaper)

Disadvantages:- ① Waste of power is high.

② Efficiency is very low, because the power is dissipated in the form of heat even i/p is absent.

Class B power amplifier:- It is an amplifier in which the transistor is biased such that, the o/p current flows for one half cycle (180°) of the i/p signal and this is achieved by setting Q point at cut-off region of load line.



* Consider the transistor is biased in cut-off region, hence transistor conducts during +ve half cycle of i/p and collector current flows. During -ve half cycle of i/p

O/p current is zero.

* When no i/p signal is applied, transistor is off, No o/p current, No power dissipation.

* When i/p signal applied, the transistor handles avg. current which increases for large i/p signals.

* The avg current in class B less than class A, hence amount of power dissipated in class B is less.

characteristics :- 1) The o/p current flows for one half cycle of i/p signal ⑧

2) Transistor dissipates no power for no i/p applied. When i/p increases

P_{dc} also increases.

3) Avg. current drawn by class B is less than class A. Hence the amount of power dissipated is less in class B. Hence overall efficiency increased.

Power relations :- $P_{in(dc)} = V_{cc} \cdot I_{dc}$
 $= V_{cc} \cdot \frac{I_m}{\pi}$

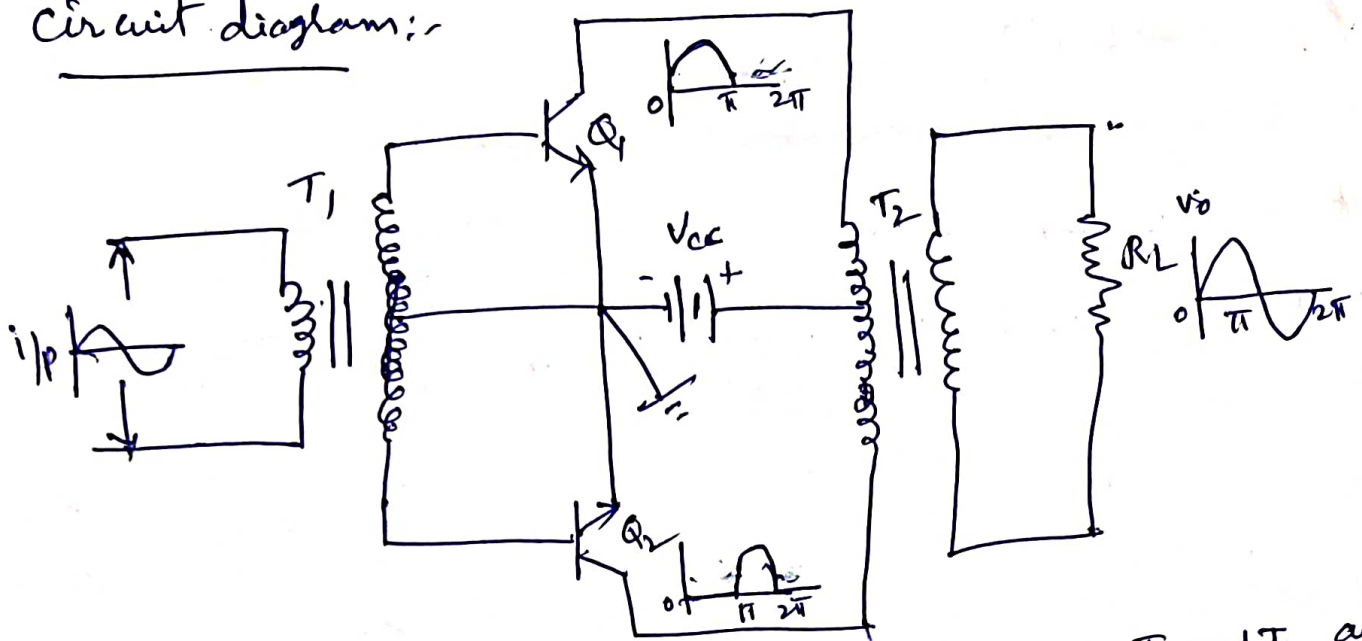
$$P_o(ac) = \frac{1}{2} (V_{rms})(I_{rms})$$
$$= \frac{1}{2} \left(\frac{V_{cc}}{\sqrt{2}} \right) \left(\frac{I_m}{\sqrt{2}} \right) = \frac{1}{4} V_{cc} I_m$$

$$\therefore \% \eta = \frac{P_o(ac)}{P_i(dc)} \times 100 = \frac{\frac{1}{4} V_{cc} I_m}{\frac{V_{cc} \cdot I_m}{\pi}} \times 100 = \frac{\pi}{4} = 78.5\%$$

Class B push pull Amplifier

Need for push pull configuration :- In class B configuration, the o/p current flows for only one half cycle for i/p. So either +ve or -ve half cycle is missing in o/p which produces large distortion. To avoid this push pull configuration is needed. In this case each transistor conducts during one half cycle and other transistor during next half cycle of i/p.

Circuit diagram:-



* It consists of two centre tapped transformers T_1 and T_2 and two identical transistors Q_1 and Q_2 .

* T_1 is i/p transformer and is also called as phase shifter. This produces two signals voltages which are 180° out of phase each other.

* These two signals with opposite polarity drives the i/p of transistors Q_1 and Q_2 .

* T_2 is o/p transformer couples ac output to load.

Operation:- * The Transistors Q_1 and Q_2 are biased at cut off by keeping zero base ... on each transistor bias which is achieved by connecting emitter and base terminals together. To have balanced circuit, two emitters are connected to the centre tap of T_1 secondary and V_{cc} to centre tap of T_2 secondary.

* When no i/p signal Q_1 and Q_2 are cut off, hence no o/p current and no power dissipation due to both the transistors.

* If transistor operated above cut-off point, transformers operate in zero flux. This is because of the transistor bias currents flows in opposite direction. (10)

* When the +ve half cycle of i/p signal is applied, Q_1 base becomes +ve, Q_1 ON and Q_2 base is -ve and Q_2 is off.

* When the -ve half cycle is applied, Q_1 is off and Q_2 is ON, so at any instant only one transistor is conducting.

* The o/p transformer combines these two o/p's and produces a full sine wave in the load resistor.

Advantages:- (1) The efficiency is higher than class A operation.

(2) If the i/p is absent, there is no power dissipation.

(3) Due to push pull connection all even harmonics are cancelled and it reduces the harmonic distortion.

(4) Due to transformer coupling, it provides proper impedance matching.

Disadvantages:- (1) The circuit is bulky and costly.

(2) Frequency response is poor.

Power relations:- $P_{dc} = \frac{V_{cc} I_m}{\pi}$ per transistor

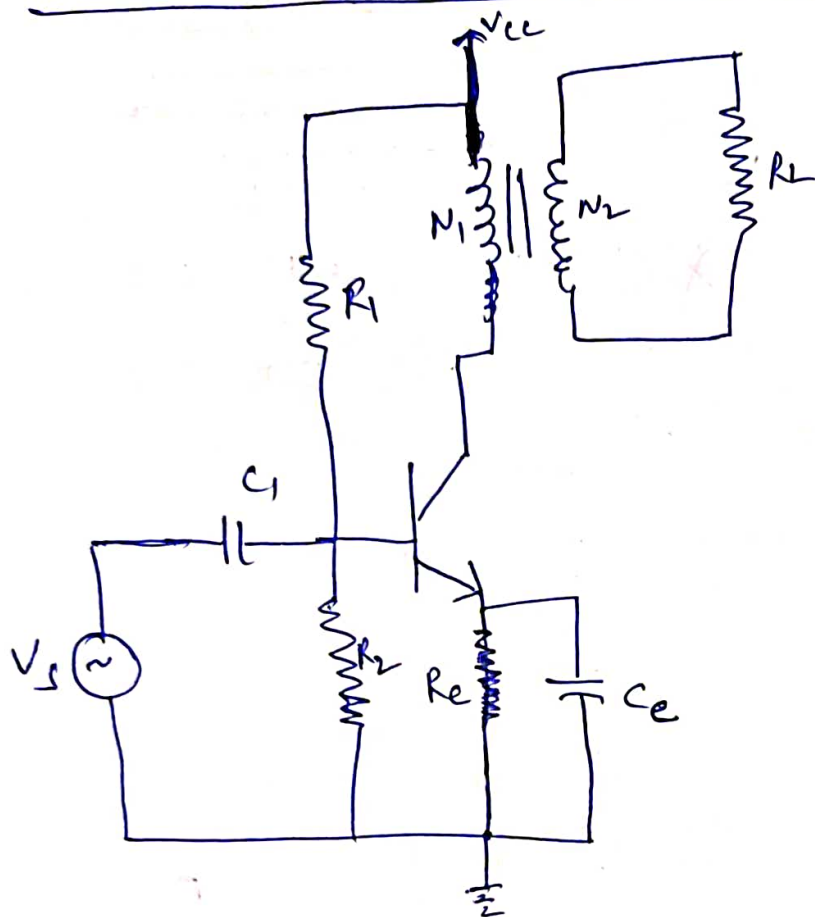
(i/p) $P_{dc} = \frac{2 V_{cc} I_m}{\pi}$ for both transistors

$$(o/p) P_{ac} = \frac{V_m \cdot I_m}{\sqrt{2} \cdot \sqrt{2}} = \frac{V_m I_m}{2} = \frac{V_{cc} I_m}{2}$$

$$\% \eta = \frac{\frac{V_{cc} I_m}{2}}{\frac{2 V_{cc} I_m}{\pi}} \times 100$$

$$= \frac{\pi}{4} \times 100 = \underline{\underline{78.5\%}}$$

Transformer coupled class A Amplifier (b1) Single ended class A Amplifier (11)



(i) Due to R_L is directly coupled to o/p circuit of amplifier the overall efficiency of direct coupled class A amplifier does not exceed 25%.

(ii) Because the I_{CQ} passes through the resistance, there is waste of power and this problem is solved by using a transformer as a coupling device between o/p of transistor and load.

Since the primary winding of transformer has a low resistance, so the power absorbed in the winding is negligible as compared to resistive load.

The function of the transformer is to match the low impedance load to high o/p impedance of amplifier.

The impedance matching property follow from the relation

$$V_1 = \frac{N_1}{N_2} \cdot V_2 \quad \text{and} \quad I_1 = \frac{N_2}{N_1} \cdot I_2$$

V_1 = primary voltage

V_2 = Secondary voltage

N_1 = No. of turns in primary

I_1 = primary current

I_2 = Secondary current

N_2 = NO. of turns in secondary

Transformer ratio $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = \frac{1}{n}$

(12)

In practical transformer, the load resistance due to transformer is assumed to be zero ohms and no dc voltage drops across the transformer. As a result, the dc load line is vertical straight line passing through Q point, thus $V_{CEQ} = V_{CC}$.

So total power drawn from dc power supply is equal to dc power developed within this transistor.

Power Relation for class A transformer coupled amplifier

$$\text{Avg. power applied to circuit} = P_{in}(dc) = V_{CC} \cdot I_{CQ}$$

$$\text{power developed in transistor } P_{TQ}(dc) = V_{CEQ} \cdot I_{CQ} = V_{CC} \cdot I_{CQ}$$

$$\text{peak value of o/p voltage across load} = V_p = V_{CEQ}$$

$$\text{and peak value of load current } I_p = I_{CQ}$$

$$\eta_o = \frac{P_o(ac)}{P_{in}(dc)} = \frac{V_p I_p / 2}{V_{CEQ} I_{CQ}} = \frac{V_m I_m}{2 V_{CEQ} I_{CQ}}$$

$$\text{because of transformer coupling, } V_{CC} = V_{CEQ} = V_m$$

$$\therefore \eta_o = \frac{V_{CEQ} \cdot I_{CQ}}{2 V_{CEQ} \cdot I_{CQ}} \times 100 = \underline{50\%}$$

$$\text{The collector efficiency} = \frac{P_{ac}}{P_{i(dc)}} \times 100 = \frac{V_p I_p / 2}{V_{CC} I_{CQ}} \times 100 = \underline{50\%}$$

Adv:- (1) The efficiency of amplifier is increased from 25% to 50%

(2) The load is connected through transformer, then the collector power dissipation is less.

Disadv:- (1) Due to transformer, the circuit becomes bulky and costlier.

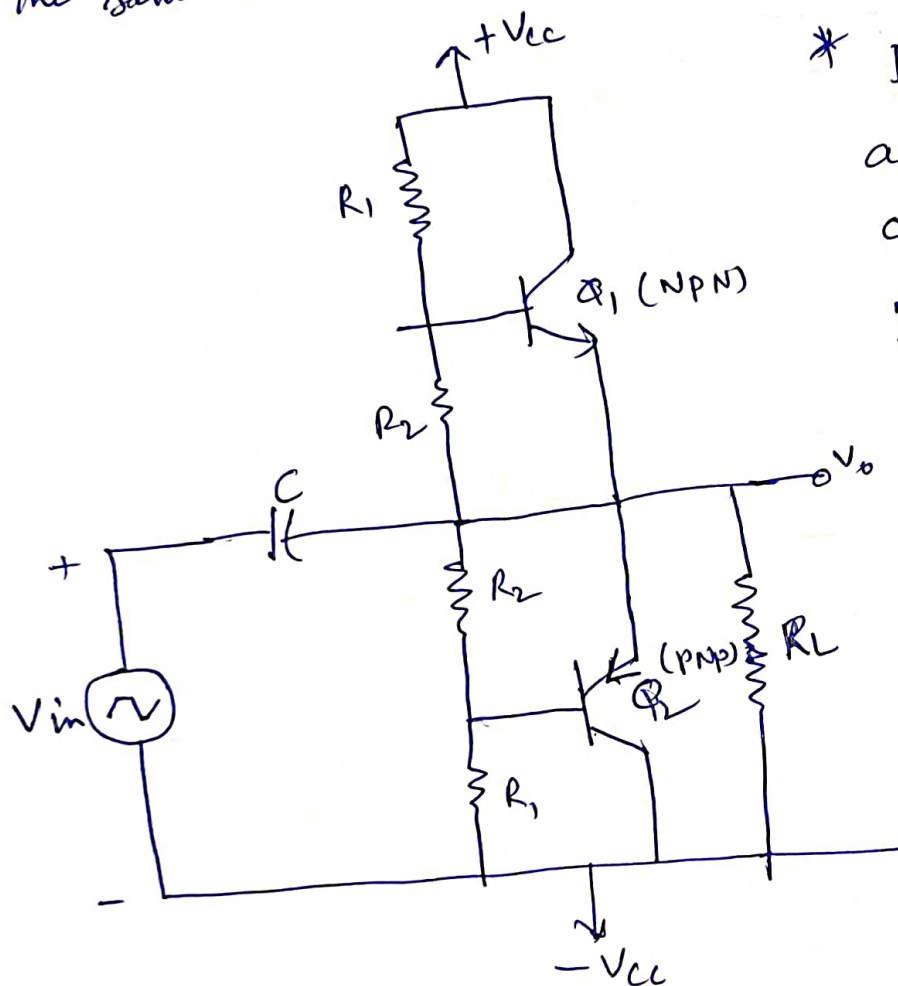
Complementary Symmetry class B Amplifier :-

(13)

Complementary means — two identical transistors but one NPN & other PNP

Symmetry means — biasing resistors connected in both transistors are equal.

As a result of this emitter base junction of each transistor is biased with the same voltage.



* During the + half cycle of ac i/p NPN transistor (Q_1) ON, PNP transistor (Q_2) OFF. During this process +ve half cycle current flows through R_L .

* During the -ve half cycle of ac i/p NPN transistor (Q_1) OFF, PNP transistor (Q_2) ON and the -ve half cycle current flows through R_L .

* Therefore when the o/p is tapped across R_L and we get a complete amplified waveform of the i/p signal.

* The amplifier has unity gain because of the emitter follower configuration.

* The split supply used in the circuit so that the dc component of V_o can be made zero. Thus only ac component of the power is available at the o/p and no need of coupling capacitor at the o/p.

Power Relations

(14)

$$P_{dc} = P_{dc}(Q_{1ON}) + P_{dc}(Q_{2ON}) = \frac{V_{CC}}{2} \cdot I_1 + \frac{V_{CC}}{2} I_2 \\ = \frac{V_{CC}}{2} (I_1 + I_2)$$

$$\text{Let } I_1 = I_2 = \frac{I_m}{\pi}$$

$$\therefore P_{dc} = \frac{V_{CC}}{2} \left(\frac{2I_m}{\pi} \right) = \frac{V_{CC} I_m}{\pi} = \frac{V_{CC} V_{CC}}{2\pi R_L} = \frac{V_{CC}^2}{2\pi R_L}$$

$$P_{ac} = P_{ac}(Q_{1ON}) + P_{ac}(Q_{2ON})$$

$$= I_{1rms}^2 R_L + I_{2rms}^2 R_L$$

$$\text{But, } I_{1rms} = I_{2rms} = \frac{I_m}{\sqrt{2}}$$

$$\therefore P_{ac} = \frac{I_m^2 R_L}{2} + \frac{I_m^2 R_L}{2} = I_m^2 R_L = \frac{V_{CC}^2}{8R_L}$$

$$\left[\because I_m = \frac{V_{CC}}{2\sqrt{2}R_L} \right]$$

$$\therefore \eta = \frac{P_{ac}}{P_{dc}} \times 100 = \frac{\frac{V_{CC}^2}{8R_L}}{\frac{V_{CC}^2}{2\pi R_L}} \times 100 = \frac{\pi}{4} \times 100 \\ = \underline{\underline{78.5\%}}$$

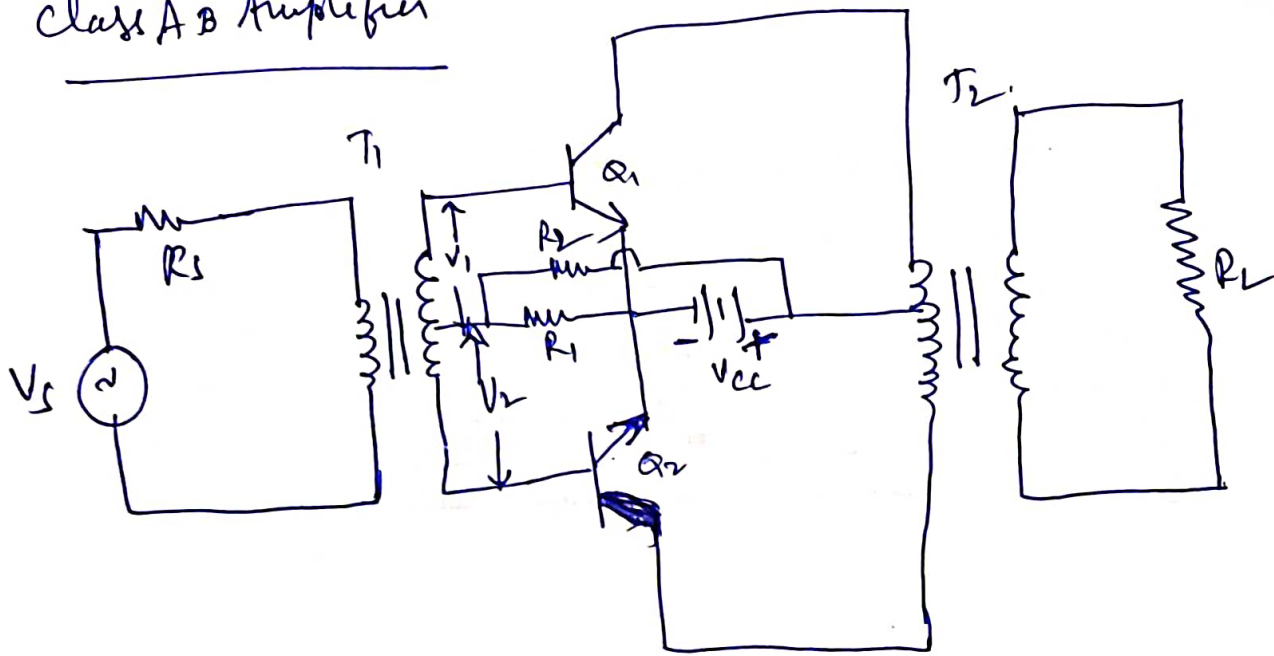
Adv:- (1) Due to absence of transformer, the weight, size and cost are less.

(2) Due to collector configuration, impedance matching is possible.

Disadv:- (1) produces cross over distortion.

Class AB Amplifier

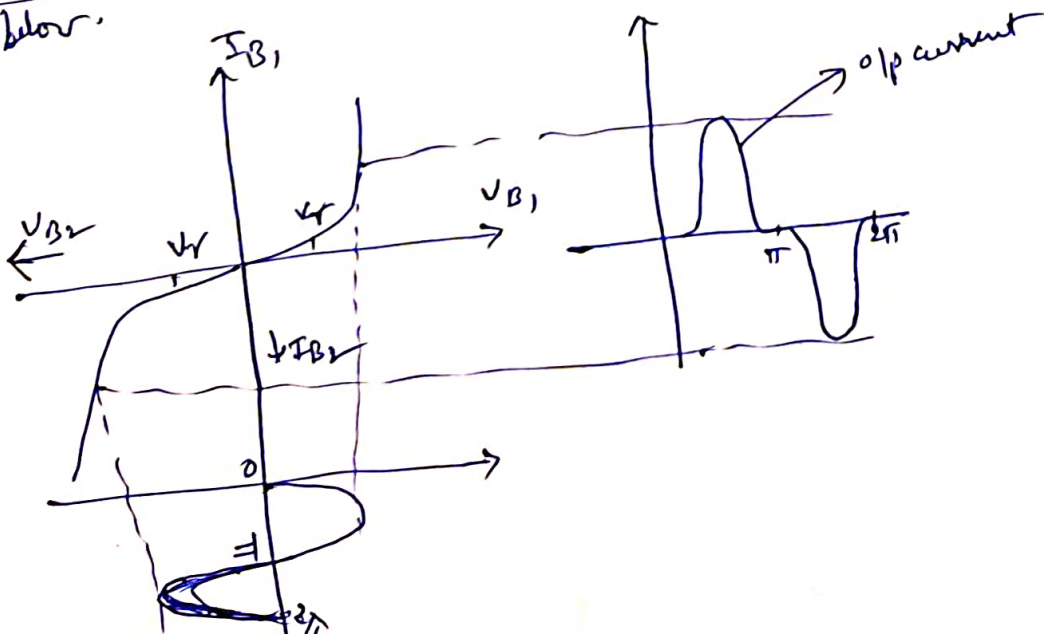
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Operation :- * The voltage drop across R_2 is so adjusted that the collector current flows for less than one cycle but more than half cycle of i/p signal. Thus the distortion in class AB amplifier is less than class B amplifier.

Class AB operation is particularly suitable for minimising crossover distortion caused by non linearity of i/p characteristics.

No appreciable base current flows until the emitter junction is forward-biased by cut in voltage V_T . Under these circumstances a sinusoidal base-voltage excitation will not result in a sinusoidal o/p current. The crossover distortion caused by non linear transistor i/p characteristic is indicated in the fig below.



* O/p is proportional to the base current.

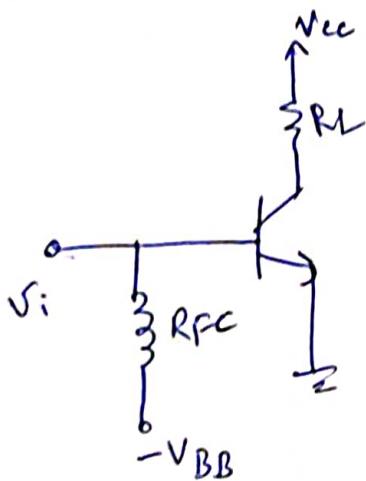
(16)

* O/p current is small when $V_i < V_T$ and thus the O/p is deviated from the linear characteristics. This distortion is called "crossover distortion".

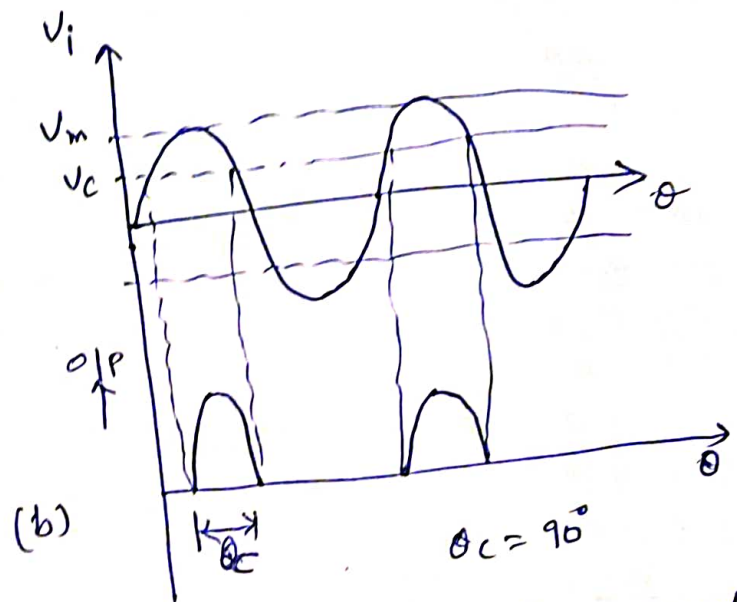
* To minimize this distortion, a small stand by current flows at zero excitation. The voltage drop across R_1 is adjusted to approximately equal to V_T ...

* class AB operation results in less distortion than class B.

Class C power Amplifier



(a)



(b)

* As per definition of class C mode of operation, it is an amplifier whose O/p conducts load current less than one half cycle of i/p .

* Fig (a) shows a simple class C amplifier with resistive load.

* The base of transistor is biased by $-V_{BB}$ connected through RFC (Radio frequency choke). It represents a high i/p impedance to high frequency i/p and it acts as short circuit and thereby the dc source prevents from short circuiting the ac i/p .

* In order to start conducting the transistor, the i/p signal (17) must overcome the $-V_{BB}$ and V_{BE} drop.

ie, $V_c = V_{BB} + 0.7 = \text{voltage at which transistor conducts.}$
 The transistor is cut off until $V_{in} < V_c$ and starts conducting when $V_{in} > V_c$ as shown in fig(b).

If V_{BB} is more than, shortens the conduction interval.
 The conduction angle is given by $\theta_c = 2\arccos\left(\frac{V_c}{V_m}\right)$

$V_m = \text{max. i/p voltage}$

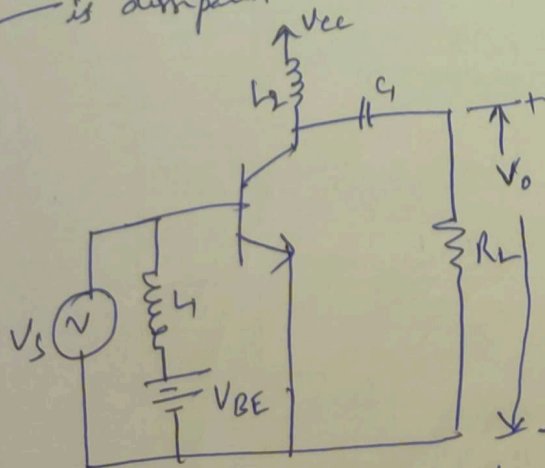
if $V_m = V_c \Rightarrow \theta_c = 0$

if $V_{BB} = 0$, $V_c = 0.7$ and $\left(\frac{V_c}{V_m}\right) \approx 0$

$\theta_c = 180^\circ$ which corresponds to class B operation.

* The amplitude of fundamental component of class C depends on the conduction angle. The greater the conduction angle, greater the amplitude of fundamental component.

* The efficiency of class C amplifier is large because very little power is dissipated when transistor is cut off.



* more load power than class A and class B amplifiers.

* Transistor is biased well beyond cutoff and it conducts only for short periods, the peak power dissipated in the collector is made to extremely large.

* The parallel LC tuned circuit across load resistor to remove all harmonics.